

The Iran War and Global Markets: A Plain-Language Summary

Version 2 — updated with the professor's additional guidance

What this project measures: how much the Iran war has actually moved markets in 2026, using a method that turns news headlines into a number a computer can measure against — and this update also asks a harder question: is that the *best* way to do it, or are there better alternatives?

The problem we're solving

Everyone agrees that news about the Iran war moves markets — oil jumps when there's a strike, stocks wobble when there's a ceasefire scare. But "war risk" isn't a number you can look up anywhere. So how do you put a number on something nobody measures directly?

The trick, borrowed from a 2003 academic paper (Rigobon and Sack, about the lead-up to the Iraq war): you don't need to measure war risk itself. You just need to know **which days had unusually intense war news**. Watch how much *more* markets moved on those noisy days compared to calm days, and that tells you how sensitive each market is to war risk — without ever guessing whether any single day's news was "good" or "bad."

The 2003 paper picked its noisy days by hand — the authors read newspapers for ten weeks and wrote down 17 dates that felt important. Our project does that same day-picking step automatically: a computer program flags each day **1 (high war news) or 0 (not)**, based on how much news coverage there was and how one-sided or split it was. That 1/0 flag is exactly what feeds into the statistical method — this is the approach the professor specifically confirmed as the right one.

Fixing our news source

In the first version, we described what happened on the big news days using Wikipedia's "Current Events Portal." That was flagged, fairly, as not a great source — Wikipedia is an encyclopedia's own summary, not journalism. In this update, every one of the 18 selected big-news days is instead described using real reporting we checked by hand against Al Jazeera, CNN and Bloomberg, with a link to the actual article for each one. (We also tried pulling real news-wire headlines automatically through a free data feed called GDELT, which does index real outlets — but it kept refusing our requests no matter how patiently we spaced them out, so for this specific step we did the verification manually instead of pretending an unreliable automated process had done it.)

Is this even the best method? We tested that directly.

This is the main new question in this update, and we didn't just argue about it — we built the obvious alternative and ran it side by side.

The alternative: instead of the fancier "compare how much noisier markets are on big-news days" method, just do the simple, standard thing — take the same 1/0 flag and run an ordinary regression: "is the average price move different on flagged days versus not?" This is called an event study, and it's the first thing most people would try.

The result: the simple method found nothing. Zero out of eighteen markets showed a statistically meaningful difference. The fancier method (the one from the 2003 paper) found meaningful effects in 13 out of 17 markets, using the *exact same* 18 days.

Why such a big gap? Of our 18 big-news days, 8 leaned "things are getting worse" and 4 leaned "things are calming down" (6 were genuinely mixed). A simple average mixes those together — a bad day that pushes stocks down and a good day that pushes them up cancel each other out in a plain average, making it look like nothing happened. But the fancier method isn't looking at averages — it's looking at how much *more spread out* the results are on big-news days, and a swing in either direction adds to that spread the same way. That's the whole reason the 2003 authors invented this method in the first place, and now we've shown concretely why it matters here too.

So: yes, for this specific problem, the fancier method is genuinely the better choice — not because it's more sophisticated, but because we tested the simple alternative head-to-head and it came up empty.

Splitting news into three flavors, as the professor suggested

Instead of just "big news day" vs. "calm day," we also split the big-news days into three groups: **bad news** (coverage leaning toward escalation), **good news** (coverage leaning toward de-escalation), and **no news** (calm days). The professor's hypothesis was specific: bad news should push yields and oil up and stocks down; good news should do the reverse.

We checked this the simplest possible way — just comparing average price moves on each type of day — and all seven markets we checked matched the hypothesis exactly: on bad-news days, both bond yields rose, oil rose, and the "fear gauge" (VIX) rose, while stocks fell. On good-news days, several of those reversed (stocks actually rose on average, and the fear gauge fell). Simple, intuitive, and it worked.

Can news content actually predict which days will be stressful? We built a

classifier to check, and got an honest "no" — which taught us something important.

We built a small machine-learning model (a standard, simple type called logistic regression) and asked it to learn: given only the news-coverage patterns for a day, can you predict whether that day will turn out to be a high-stress day for markets overall?

The honest answer: not really. The model did worse than random guessing at this specific task. That sounds bad, but digging into *why* actually reinforces our main approach rather than undermining it: the "high market stress" days we asked it to predict were defined using a broad basket of markets (bonds, stocks, oil, credit, volatility) — and plenty of those stressful days had nothing to do with Iran at all (Fed decisions, other economic news, etc.). On those days, financial news coverage was naturally focused elsewhere, not on Iran, so of course Iran-specific news coverage doesn't predict them well. This is actually a good argument for why picking "big news days" from the news itself (rather than from overall market chaos) is the right design — it stays focused on the one specific risk we're trying to measure, instead of getting confused by every other thing that moves markets.

We separately checked: does the choice of exactly *how* you draw the "big news day" line matter for the final results? We tried three different ways to draw that line — our original hand-tuned rule, the machine-learning classifier's guess, and an unsupervised pattern-finder that never even looked at prices. **The core economic story (stocks down, credit costs up) held up under all three; only a couple of markets (the VIX and the dollar) were sensitive to exactly which line was drawn**, and those happen to be the same two markets flagged as less reliable for other reasons too.

Checking whether our word list misses new war vocabulary

The professor raised a good point: any list of "war words" built ahead of time is going to miss brand-new terminology that a specific, unfolding war invents as it goes. We checked this directly against real news sentences from our 18 selected days.

Result: over half (52.5%) of clearly war-relevant sentences scored zero matches against our hand-built word list. Real examples our list completely missed: "Natanz nuclear enrichment complex" (a facility name — our list has no idea what Natanz is), "rogue supertankers" (a phrase this specific conflict's coverage invented), and "the memorandum-of-understanding ceasefire" (a named agreement that didn't exist before this war). This is a real, quantified limitation — not something we're papering over — and the honest fix would be to have a more flexible AI model read the actual sentences in context (which understands "Natanz" is a nuclear site without needing it pre-programmed), rather than count fixed phrases. We didn't build that as a full automated step in this update, but we did the work by hand on a sample to show what it would catch.

Checking our results against what the professor expected

The professor gave us a specific prediction ahead of time, based on what happened in 2003 and what's different about 2026: oil, stock, and credit-market reactions should look similar to 2003, but **interest rates should behave the opposite way**, because the US now pumps roughly 14 million barrels of oil a day versus about 6 million back in 2003 — meaning a war-driven oil price spike now benefits US oil producers a lot more than it did back then, changing how the whole economy reacts.

Every single piece of that prediction came true in our results. Oil, stocks, and credit spreads moved the same direction they did in 2003. Interest rates, inflation expectations, and the dollar all moved the *opposite* direction from 2003, exactly as predicted — because this time the war is being priced mainly as an oil-supply shock (which pushes prices and rates up) rather than a pure "the economy is in danger" scare (which pushed rates down in 2003).

Bottom line

This update did three things: it replaced a weak source (Wikipedia) with real, checked news citations; it tested — not just claimed — that the fancier statistical method is the right choice here, by building the obvious simpler alternative and watching it fail where the fancier method succeeded; and it extended the analysis with a three-way news split that matched the professor's hypothesis perfectly, plus an honest look at where our own methods (the word list, the classifier) have real limits. The headline economic story hasn't changed: a war that threatens oil supply pushes stocks down (more outside the US than in), widens credit spreads, lifts volatility, sends money into safe currencies, and — unlike the 2003 Iraq war — pushes interest rates and inflation expectations *up* rather than down, because America's much larger oil industry changes how this kind of shock hits the economy.

This is a simplified companion to the full technical report ([REPORT.md](#)), which includes all the statistical tables, confidence levels, and methodology details, including the alternative methods and literature review, for readers who want the underlying math.